Lecture 2 (*February 19*)

Reading: Sections 2.2–2.3
Viewing: Lecture 2

The basic method for solving systems of equations is **Gaussian elimination**. For instance:

$$\begin{cases} x - y + 2z = 1 \\ -2x + 2y - 3z = -1 \\ -3x - y + 2z = -3 \end{cases} \qquad \text{or, equivalently,} \qquad \begin{bmatrix} 1 & -1 & 2 \\ -2 & 2 & -3 \\ -3 & -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ -3 \end{bmatrix} \tag{8}$$

The 3×3 matrix $A = \begin{bmatrix} 1 & -1 & 2 \\ -2 & 2 & -3 \\ -3 & -1 & 2 \end{bmatrix}$ is called the **coefficient matrix**, while $\boldsymbol{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is the unknown vector. Geometrically, the system (8) consists of three linear equations in xyz–space. Each of these equations determines a 2–dimensional plane in 3–dimensional space, so the system of equations computes the coordinates (x, y, z) of the intersection point of three planes.

Gaussian elimination is the following algorithm. Consider the **augmented matrix**:

$$\left[\begin{array}{ccc|c} \boxed{1} & -1 & 2 & 1 \\ \boxed{-2} & 2 & -3 & -1 \\ \boxed{-3} & -1 & 2 & -3 \end{array}\right] \tag{9}$$

obtained by tacking the right-hand side of (8) onto the right of the matrix A. The **pivot** of each row (marked by a box) refers to the leftmost non-zero entry on that row. The game is the following: by performing **row eliminations** (namely adding any multiple of row j to row i, for various $i > j$), your goal is to ensure that the pivot of the first row is strictly to the left of the pivot on the second row, which in turn is strictly to the left of the pivot on the third row etc.

The particular row eliminations you have to do are imposed upon you by the following algorithm. Start from the pivot on the first row. I claim that there's a single choice of row eliminations which will ensure that the entries below the pivot on the first row will be 0. This is because:

$$a \cdot 1 + (-2) = 0 \text{ requires } a = 2$$
$$b \cdot 1 + (-3) = 0 \text{ requires } b = 3$$

so we must add 2 times the first row to the second row and 3 times the first row to the third row, in order to achieve our stated goal of having the entries below the first pivot be 0.

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ -2+2\cdot 1 & 2+2\cdot(-1) & -3+2\cdot 2 & -1+2\cdot 1 \\ -3+3\cdot 1 & -1+3\cdot(-1) & 2+3\cdot 2 & -3+3\cdot 1 \end{array}\right] = \left[\begin{array}{ccc|c} \boxed{1} & -1 & 2 & 1 \\ 0 & 0 & \boxed{1} & 1 \\ 0 & \boxed{-4} & 8 & 0 \end{array}\right]$$

The next step is to find the the pivot of the second row, and repeat the process. But wait! In the example above, the pivot on the second row is to the right of the pivot on the third row. Whenever this happens, there's no row elimination which will put the pivots in the right order (namely the pivot on the second row to the left of the pivot on the third row). So we allow ourselves to make an

extra move: **row exchanges**. Specifically, we simply swap the second and third rows. We obtain:

$$\left[\begin{array}{ccc|c} \boxed{1} & -1 & 2 & 1 \\ 0 & \boxed{-4} & 8 & 0 \\ 0 & 0 & \boxed{1} & 1 \end{array}\right] \tag{10}$$

This augmented matrix is in **row echelon form**, which is the purpose of Gaussian elimination: <u>all the pivots are non-zero, and each pivot is strictly to the right of the one on the row above</u>.

Remark. *We only allow the pivots to be in the original part of the matrix, not in the augmented part (i.e. not in the right-most column of* (9)*). If at any point during Gaussian elimination, some i–th row of the matrix does not have a pivot (which can only happen if that row has all zeroes except maybe on the last column) then we use a row echange to swap the i–th row with the last row. Thus, at the end of Gaussian elimination, all rows without pivots (if any) will be at the bottom of the matrix.*

Remark. *There's a variant of this algorithm, called Gauss-Jordan elimination, which entails doing two additional steps. Firstly, multiply all the rows of the augmented matrix by appropriate constants, so that <u>all the pivots are equal to 1</u>:*

$$\left[\begin{array}{ccc|c} \boxed{1} & -1 & 2 & 1 \\ 0 & \boxed{1} & -2 & 0 \\ 0 & 0 & \boxed{1} & 1 \end{array}\right]$$

Then, we do some further row operations so that <u>all the entries above the pivots are 0</u> as well. In the case of the matrix at hand, we add 2 *times the third row to the second row:*

$$\left[\begin{array}{ccc|c} \boxed{1} & -1 & 2 & 1 \\ 0 & \boxed{1} & 0 & 2 \\ 0 & 0 & \boxed{1} & 1 \end{array}\right]$$

and then we add -2 *times the third row and* 1 *times the second row to the first row:*

$$\left[\begin{array}{ccc|c} \boxed{1} & 0 & 0 & 1 \\ 0 & \boxed{1} & 0 & 2 \\ 0 & 0 & \boxed{1} & 1 \end{array}\right]$$

*The output of Gauss-Jordan elimination is said to be in **reduced row echelon form**, i.e. a matrix which is both in row echelon form, and satisfies the two extra conditions underlined above. We will revisit this notion in a few weeks' time.*

The augmented matrix (10) corresponds to the following system of linear equations:

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & -4 & 8 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \text{or, equivalently,} \quad \begin{cases} x - y + 2z = 1 \\ -4y + 8z = 0 \\ z = 1 \end{cases} \tag{11}$$

and the great thing about Gaussian elimination is that this system of equations is actually equivalent to the system (8) that we started from:

$$\boxed{\textbf{any solution } (x, y, z) \textbf{ of } (8) \textbf{ is also a solution of } (11)\textbf{, and vice versa}}$$

So why did we go through the trouble of Gaussian (or Gauss-Jordan) elimination? This is because systems where the coefficient matrix is in (reduced) row echelon form are really easy to solve by hand, by the following procedure (for which the technical term is **back substitution**): the third equation in (11) gives you $z = 1$, which you can plug into the second equation to get $y = 2$, which you can plug into the first equation to get $x = 1$. So the solution is $(x, y, z) = (1, 2, 1)$.

Remark. *If the row echelon form of a matrix A has a full row of zeroes (i.e. no pivot on the last row), then the matrix A is called* ***singular****. In this case, the system* (11) *may not have any solutions or may have more solutions than expected. Otherwise, the matrix is called* ***non-singular****.*

So what is going on behind this method? Back substitution provided such a quick answer because:

$$\boxed{U\boldsymbol{v} = \boldsymbol{c}} \tag{12}$$

is a very easy system to solve if U is in row echelon form. Indeed, you solve for the last component of the vector $\boldsymbol{v}$ first, then for the next-to-last one, then for the next-to-next-to-last one etc. If you need to solve a system of equations $A\boldsymbol{v} = \boldsymbol{b}$ for a general matrix A, then Gaussian elimination precisely provides a recipe for putting A in row echelon form, at the cost of performing the following row operations (for any $i > j$):

- Adding the j–th row times λ to the i–th row of A, which is achieved by multiplying the latter on the left with an **elimination matrix**:

$$A \rightsquigarrow E_{ij}^{(\lambda)} A \qquad \text{where} \qquad E_{ij}^{(\lambda)} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & \lambda & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

  The entry λ is on the i–th row and j–th column (in the formula above, $i = 4$ and $j = 2$). [2]

- Switching the i–th and j–th rows of A is achieved by multiplying the latter on the left with a **permutation matrix**:

$$A \rightsquigarrow P_{ij} A \qquad \text{where} \qquad P_{ij} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

  Explicitly, P_{ij} has coefficients 1 on positions (i, j), (j, i) and (k, k) for all $k \neq i, j$, and all other coefficients of P_{ij} are 0 (in the formula above, $i = 3$ and $j = 4$).

[2] If you're doing Gauss-Jordan elimination, you will also need the elimination matrices $E_{ij}^{(\lambda)}$ for $i < j$.

- If you wish to put the matrix in reduced row echelon (as opposed to just row echelon) form, you will also need to multiply the i–th row of A by various constants λ, which is achieved by multiplying A on the left by a **diagonal** matrix:

$$A \rightsquigarrow D_i^{(\lambda)} A \qquad \text{where} \qquad D_i^{(\lambda)} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \lambda & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

  The entry λ is on the i–th row (in the formula above, $i = 2$).

So let's see what these matrices amount to in the example of (8). We will forget about the 3×4 augmented matrix $[A|\boldsymbol{b}]$ and just discuss Gaussian elimination on the 3×3 matrix:

$$A = \begin{bmatrix} 1 & -1 & 2 \\ -2 & 2 & -3 \\ -3 & -1 & 2 \end{bmatrix}$$

The first thing we did was to add 2 times the first row to the second row:

$$E_{21}^{(2)} A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \\ -2 & 2 & -3 \\ -3 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 0 & 1 \\ -3 & -1 & 2 \end{bmatrix}$$

The next thing we did was to add 3 times the first row to the third row:

$$E_{31}^{(3)} E_{21}^{(2)} A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \\ 0 & 0 & 1 \\ -3 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 0 & 1 \\ 0 & -4 & 8 \end{bmatrix}$$

Then we exchanged the second and third rows:

$$P_{23} E_{31}^{(3)} E_{21}^{(2)} A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \\ 0 & 0 & 1 \\ 0 & -4 & 8 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & -4 & 8 \\ 0 & 0 & 1 \end{bmatrix} =: U \tag{13}$$

and we have achieved precisely the row echelon form matrix of (11).

Remark. *Suppose we wish to go further, and do Gauss-Jordan elimination. Then we need to make all the pivots 1, and to this end we must multiply the second row by $-1/4$:*

$$D_2^{\left(-\frac{1}{4}\right)} U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{4} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \\ 0 & -4 & 8 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

The pivots are the 1's on the diagonal. To put U in reduced row echelon form, we need to annihilate all the non-zero entries above the 1's. This is achieved by adding 2 times the third row to the second row, then adding -2 times the third row to the first row, then adding the second row to the first:

$$E_{12}^{(1)} E_{13}^{(-2)} E_{23}^{(2)} D_2^{\left(-\frac{1}{4}\right)} U = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} =: R$$

The matrix R is now in reduced row echelon form.

Very importantly, this procedure also allows us to convert the system of equations (8) into a system of the form (12). Specifically, take the equality (8) and multiply it on the left with the particular product of elimination/permutation/diagonal matrices which appears in (13). We obtain:

$$\underbrace{P_{23}E_{31}^{(3)}E_{21}^{(2)}A}_{U}\boldsymbol{v} = P_{23}E_{31}^{(3)}E_{21}^{(2)}\boldsymbol{b} \tag{14}$$

If we let:

$$\boldsymbol{c} := P_{23}E_{31}^{(3)}E_{21}^{(2)}\boldsymbol{b} = \begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}\begin{bmatrix}1&0&0\\0&1&0\\3&0&1\end{bmatrix}\begin{bmatrix}1&0&0\\2&1&0\\0&0&1\end{bmatrix}\begin{bmatrix}1\\-1\\-3\end{bmatrix} =$$

$$= \begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}\begin{bmatrix}1&0&0\\0&1&0\\3&0&1\end{bmatrix}\begin{bmatrix}1\\1\\-3\end{bmatrix} = \begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}\begin{bmatrix}1\\1\\0\end{bmatrix} = \begin{bmatrix}1\\0\\1\end{bmatrix}$$

then formula (14) reads precisely (12), which establishes the following important principle:

Fact 3. *If:*

$$\boxed{A = LU \text{ and } \boldsymbol{b} = L\boldsymbol{c}}$$

for some matrix L (which in practice will be a product of elimination/permutation/diagonal matrices), then the system $A\boldsymbol{v} = \boldsymbol{b}$ is equivalent (i.e. has the same solution $\boldsymbol{v}$) as the system $U\boldsymbol{v} = \boldsymbol{c}$.

Since we have already seen that it is easy to solve a system of equations where the matrix U is in row echelon form, this gives a pretty general algorithm for solving general systems of equations.

Remark. *Gaussian (or Gauss-Jordan) elimination can be applied to any $m \times n$ matrix, producing a matrix in row (or reduced row) echelon form. The fact that in the first part of this lecture, we applied it to the augmented matrix $[A|\boldsymbol{b}]$ was simply because this is what you're supposed to do if you want Gaussian elimination to help you solve systems of equations. But as we have seen in the second part of this lecture, the method of Gaussian elimination can be applied to the matrix A itself.*